

Section 1

Create database hive_test

```
CREATE DATABASE hive_test; USE hive_test;
```

Create table statement

```
CREATE TABLE salesman ( salesman_id INT, name STRING, city STRING, commission DOUBLE  
) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' STORED AS TEXTFILE;
```

```
CREATE TABLE customer ( customer_id INT, cust_name STRING, city STRING, grade INT, salesman_id  
INT ) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' STORED AS TEXTFILE;
```

```
CREATE TABLE orders ( ord_no INT, purch_amt DOUBLE, ord_date DATE, customer_id INT, salesman_id  
INT ) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' STORED AS TEXTFILE;
```

```
load data local inpath '/home/takeo/salesman.csv' into table salesman;
```

```
load data local inpath '/home/takeo/customer.csv' into table customer;
```

```
load data local inpath '/home/takeo/order.csv' into table orders;
```

Write a SQL statement to prepare a list with salesman name, customer name, and their cities for the salesmen and customer who belongs to the same city.

```
select s.name as salesman_name, c.cust_name as customer_name, s.city as city from salesman s join  
customer c on s.city=c.city;
```

Write a SQL statement to know which salesman are working for which customer.

```
select s.name as salesman_name, c.cust_name as customer_name from salesman s join customer c on  
s.salesman_id=c.salesman_id;
```

Write a SQL statement to make a list with order no, purchase amount, customer name and their cities for those orders which order amount between 500 and 2000.

```
select o.ord_no, o.purch_amt, c.cust_name, c.city from orders o join customer c on o.customer_id =  
c.customer_id where o.purch_amt between 500 and 2000;
```

Write a SQL statement to find the list of customers who appointed a salesman for their jobs who gets a commission from the company is more than 12%.

```
select c.cust_name, s.name as salesman_name, s.commission from customer c join salesman s on  
c.salesman_id = s.salesman_id where s.commission > 0.12;
```

Write a SQL statement to find the list of customers who appointed a salesman for their jobs who does not live in the same city where their customer lives, and gets a commission above 12% .

```
select s.name as salesman_name, c.cust_name, s.city as salesman_city, s.commission, c.city as customer_city from customer c join salesman s on c.salesman_id = s.salesman_id where s.commission > 0.12 and s.city <> c.city;
```

Section 2

Query 1: Create table with complex data

```
create table salesdetail_complex (Product_ID INT,productdetails map<String,String>,Order_Priority VARCHAR(4),merchantType CHAR(4),Sale_Amount DOUBLE,Order_Quantity BIGINT,Discount FLOAT,Salaryhike TINYINT,companyprofit SMALLINT, financeDeficit DECIMAL(8,2),indian BOOLEAN,saledate array,saleyear array,selleramountfile array,orderQuantityfile array,costlist map<int,int>,strutureType struct<city:string,state:string,pin:bigint>,systemdatetime array) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' collection items terminated by '$' map keys terminated by '#';
```

```
nano salesdetail.txt
```

```
load data local inpath '/home/takeo/salesdetail.txt' into table salesdetail_complex;
```

Query 2: view 2 records

```
select * from salesdetail_complex limit 2;
```

Query 3: Non partition table

```
CREATE TABLE non_part ( dateid SMALLINT, caldate DATE, day STRING, week SMALLINT, month STRING, qtr STRING, year SMALLINT, holiday BOOLEAN ) ROW FORMAT DELIMITED FIELDS TERMINATED BY '|' STORED AS TEXTFILE;
```

```
load data local inpath '/home/takeo/non_part.txt' into table non_part;
```

Query 4: Partition table on caldate, load from non partition

```
CREATE TABLE date_part ( dateid SMALLINT, day STRING, week SMALLINT, month STRING, qtr STRING, year SMALLINT, holiday BOOLEAN ) PARTITIONED BY ( caldate DATE ) STORED AS TEXTFILE;
```

```
SET hive.exec.dynamic.partition=true; SET hive.exec.dynamic.partition.mode=nonstrict;
```

```
INSERT INTO TABLE date_part PARTITION (caldate) SELECT dateid, day, week, month, qtr, year, holiday, caldate FROM non_part;
```

```
show partitions date_part;
```

Query 5: Create a partitioned and bucketed table

```
CREATE TABLE date_part_bucket ( dateid SMALLINT, day STRING, week SMALLINT, month STRING, qtr STRING, year SMALLINT, holiday BOOLEAN ) PARTITIONED BY ( caldate DATE ) CLUSTERED BY (dateid) INTO 4 BUCKETS STORED AS ORC;
```

```
SET hive.exec.dynamic.partition=true; SET hive.exec.dynamic.partition.mode=nonstrict;
```

```
SET hive.enforce.bucketing=true;
```

```
INSERT INTO TABLE date_part_bucket PARTITION (caldate) SELECT dateid, day, week, month, qtr, year,  
holiday, caldate FROM non_part;
```

```
SELECT * FROM date_part_bucket LIMIT 5;
```